

AI-Powered Enterprise E-Commerce Platform

AI Model Architecture Specification

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1.0	DD/MM/YYYY	Initial Version

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1. Introduction

Artificial Intelligence is a major component of this platform.

The AI subsystem provides intelligent capabilities including:

- Personalized recommendations
- Fraud detection
- Sales forecasting
- Customer segmentation
- Conversational assistance

The architecture is designed to be modular, scalable, and deployable as independent microservices.

2. Purpose

This document aims to:

- Define AI components.
 - Specify model architectures.
 - Define training pipelines.
 - Establish deployment strategies.
 - Support future model improvements.
-

3. AI Objectives

AI-01

Improve customer experience.

AI-02

Increase sales through recommendations.

AI-03

Detect fraudulent activities.

AI-04

Provide business intelligence.

AI-05

Enable intelligent customer interactions.

4. High-Level AI Architecture



5. AI Services Overview

Service	Purpose
Recommendation Service	Product recommendations
Fraud Service	Fraud risk detection
Forecast Service	Sales forecasting
Segmentation Service	Customer grouping
AI Assistant	Conversational support

6. AI Data Architecture

Data Sources

User Data

Examples:

- User profile
 - Purchase history
 - Browsing behavior
-

Product Data

Examples:

- Categories
 - Ratings
 - Prices
-

Transaction Data

Examples:

- Orders
 - Payment history
 - Fraud labels
-

Historical Business Data

Examples:

- Sales history
 - Seasonal trends
-

Data Flow

```
Raw Data
  ↓
Cleaning
  ↓
Feature Engineering
  ↓
Training Dataset
```

↓
Models

7. Recommendation System Architecture

Business Objective

Recommend relevant products to users.

Candidate Algorithms

Phase 1

Popularity-Based Recommendations

Phase 2

Content-Based Filtering

Phase 3

Collaborative Filtering

Phase 4 (Future)

Hybrid Recommendation System

Recommended Initial Approach

Hybrid:

Content-Based
+
Collaborative Filtering

Inputs

- User history
- Product metadata
- Ratings
- Purchase history

Outputs

Recommended product list.

Pipeline

```
graph TD;
    A[Customer Activity] --> B[Feature Generation];
    B --> C[Recommendation Model];
    C --> D[Ranked Products];
```

Evaluation Metrics

- Precision@K
 - Recall@K
 - MAP
 - NDCG
-

8. Fraud Detection Architecture

Business Objective

Identify suspicious transactions.

Inputs

- Payment amount
- Order frequency

- Device information
- User behavior

Candidate Algorithms

- Logistic Regression
- Random Forest
- XGBoost

Recommended Model

XGBoost

Reasons:

- High performance
- Handles imbalanced data well
- Good explainability

Output

Fraud Probability Score.

Pipeline

```
Transaction Data
  ↓
Feature Engineering
  ↓
Fraud Model
  ↓
Risk Score
```

Evaluation Metrics

- Accuracy

- Precision
- Recall
- F1 Score
- ROC-AUC

9. Sales Forecasting Architecture

Business Objective

Predict future sales.

Inputs

- Historical sales
 - Promotions
 - Seasonality
 - Holidays
-

Candidate Models

- Linear Regression
 - ARIMA
 - Prophet
 - LSTM
-

Recommended Initial Model

Facebook Prophet

Reasons:

- Easy implementation
 - Handles seasonality
 - Strong baseline performance
-

Future Enhancement

LSTM Models.

Outputs

Future sales predictions.

Evaluation Metrics

- MAE
 - RMSE
 - MAPE
-

10. Customer Segmentation Architecture

Business Objective

Group similar customers.

Inputs

- Purchase frequency
 - Spending behavior
 - Product preferences
-

Feature Framework

RFM Analysis:

- Recency
 - Frequency
 - Monetary Value
-

Candidate Models

- K-Means
- DBSCAN

- Hierarchical Clustering

Recommended Model

K-Means Clustering

Outputs

Customer segments.

Examples:

- High Value Customers
- Loyal Customers
- At Risk Customers

Evaluation Metrics

- Silhouette Score
- Davies-Bouldin Index

11. AI Assistant Architecture

Business Objective

Provide conversational support.

Features

- Product questions
- Order tracking
- Recommendation assistance

Initial Implementation

Rule-Based Assistant.

Future Enhancement

RAG-Based Assistant.

Pipeline

```
graph TD; A[User Query] --> B[Intent Detection]; B --> C[Response Generation];
```

12. Feature Engineering Pipeline

Recommendation Features

- Purchase count
 - User-product interactions
 - Product categories
-

Fraud Features

- Transaction frequency
 - Device usage
 - Spending patterns
-

Forecast Features

- Time features
 - Seasonal indicators
 - Promotional indicators
-

Segmentation Features

- RFM metrics
 - Behavioral statistics
-

Pipeline

```
Raw Data
  ↓
Cleaning
  ↓
Encoding
  ↓
Scaling
  ↓
Feature Store
```

13. Model Training Pipeline

Training Workflow

```
Data Collection
  ↓
Data Cleaning
  ↓
Feature Engineering
  ↓
Training
  ↓
Evaluation
  ↓
Model Registration
```

Training Frequency

Recommendation

Weekly.

Fraud

Monthly.

Forecasting

Monthly.

Segmentation

Monthly.

14. Model Storage

Store:

- Model files
- Metadata
- Evaluation reports

Recommended structure:

```
models/  
|  
├─ recommendation/  
├─ fraud/  
├─ forecasting/  
└─ segmentation/
```

15. Model Deployment Architecture

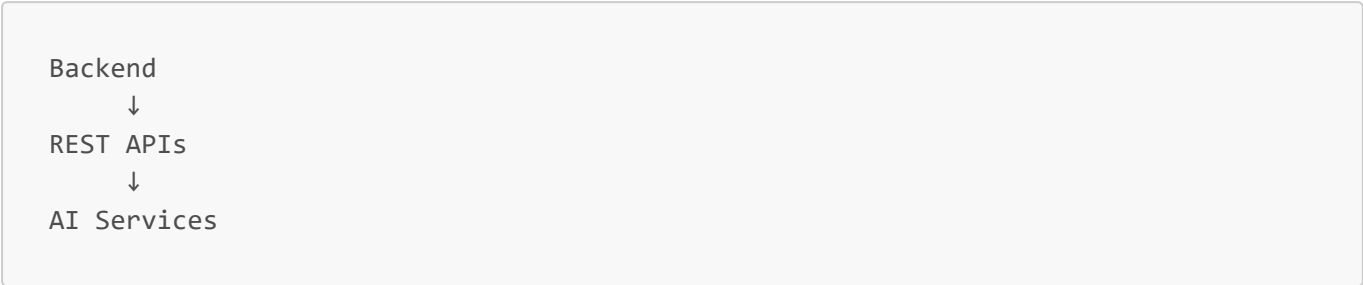
Deployment Approach

Independent AI microservices.

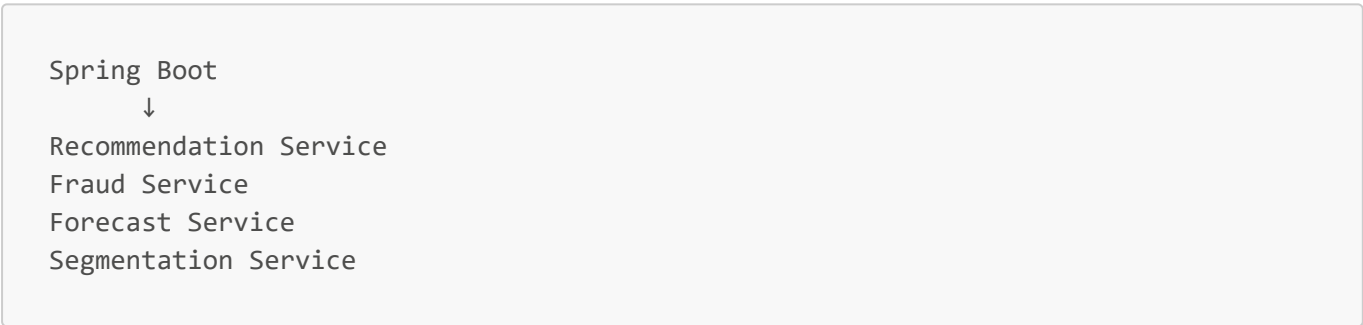
Technologies

- FastAPI
- Docker

Service Communication



Deployment Diagram



16. Model Monitoring

Monitor:

- Latency
- Error rates
- Accuracy degradation
- Data drift
- Resource usage

Metrics Examples

Metric	Purpose
Prediction Latency	Performance
Accuracy	Model Quality
Drift Score	Dataset Changes

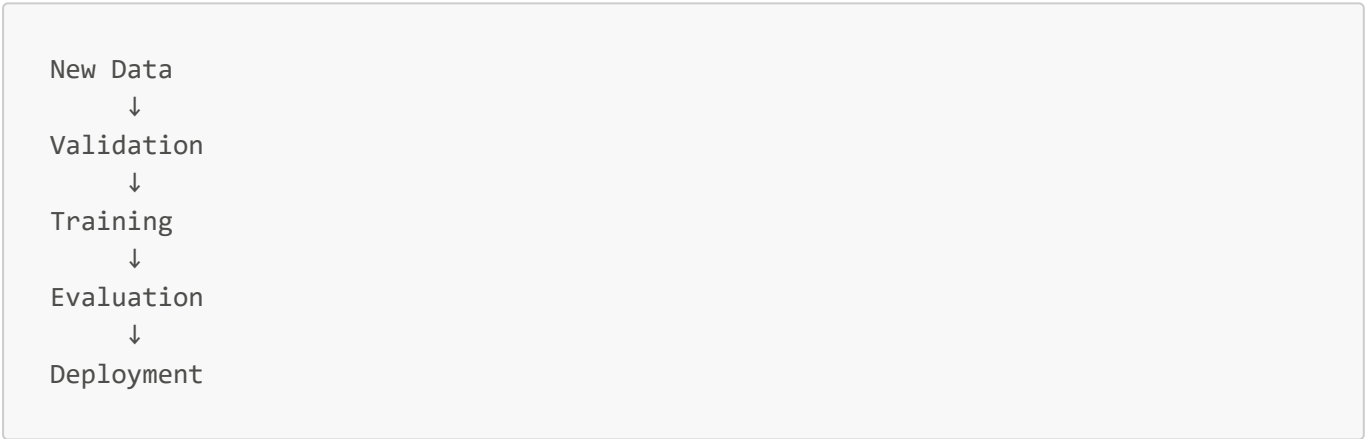
Metric	Purpose
Error Rate	Stability

17. Retraining Strategy

Triggers

- Scheduled retraining
- Accuracy degradation
- Data drift detection

Workflow



18. AI Risks and Mitigation

Risk	Mitigation
Poor Data Quality	Data Validation
Overfitting	Cross Validation
Model Drift	Monitoring
Bias	Fairness Evaluation
High Latency	Caching

19. AI Security Considerations

Protect against:

- Model extraction
- Data poisoning
- Adversarial attacks
- Unauthorized access

Security mechanisms:

- API authentication
- Dataset validation
- Monitoring
- Rate limiting

20. AI Performance Targets

Service	Target
Recommendation Latency	<5 sec
Fraud Detection Latency	<2 sec
Forecast Generation	<10 sec
Segmentation	Batch Processing

21. Future Enhancements

Recommendation

- Deep Learning Recommenders
 - Reinforcement Learning
-

Fraud Detection

- Graph Neural Networks
 - Real-Time Fraud Detection
-

Forecasting

- LSTM
 - Transformer Models
-

AI Assistant

- RAG Architecture
- LLM Integration

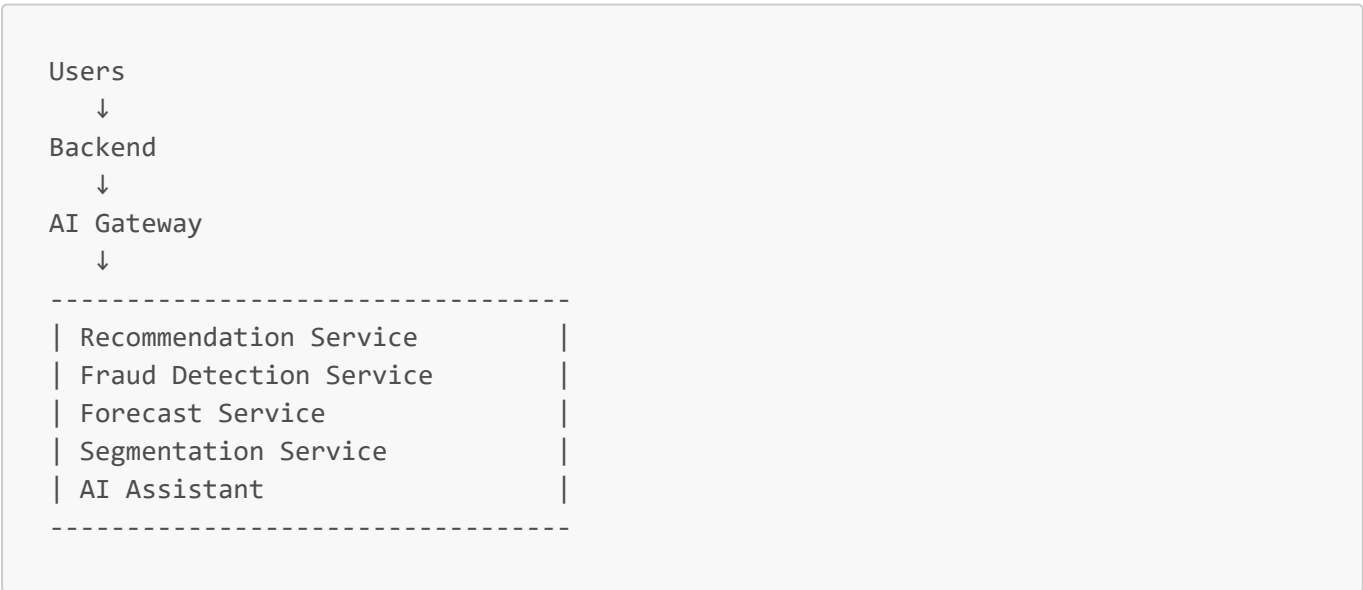
MLOps

- MLflow
- Feature Store
- Automated Retraining Pipelines

22. Recommended AI Directory Structure

```
ai-services/
|
|— recommendation/
|— fraud_detection/
|— forecasting/
|— segmentation/
|— assistant/
|
|— datasets/
|— notebooks/
|— models/
|— pipelines/
|— evaluations/
```

23. Recommended Architecture Diagram



24. Conclusion

The proposed AI architecture provides:

- modular AI services,
- scalable deployment,
- enterprise-level model management,
- future MLOps readiness,
- and strong business intelligence capabilities.

The architecture enables gradual evolution from classical machine learning solutions to advanced AI systems while maintaining maintainability and scalability.

References

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- [5] XGBoost Documentation.
- [6] Scikit-Learn Documentation.
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- [8] Google MLOps Whitepaper.